API Contract

Base URL: /api/v1

Content-Type: application/json

Authentication: Bearer JWT (except login/register)

Standard Error Responses

Unauthorized (401)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 401,
  "error": “Unauthorized",
  "message": "User not logged in",
  "path": “api path"
}
```

Internal Server Error (500)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 500,
  "error": "Internal Server Error",
  "message": "Server failed to respond",
  "path": "api path"
}
```

1. Register User

POST /auth/register

Request

```
{
  "email": "user@example.com",
  "password": "StrongPassword123",
  "name": "Ravi Kumar",
  "phoneNo": "9876543210",
  "dateOfBirth": "1995-06-12"
}
```

Responses

Success (201)

```
{
  "userId": "uuid",
  "message": "User registered successfully"
}
```

Email already exists (409)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 409,
  "error": "Conflict",
  "message": "Email already registered",
  "path": "/api/v1/auth/register"
}
```

Invalid input (400)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid registration details",
  "path": "/api/v1/auth/register"
}
```

2. Login

POST /auth/login

Request

```
{
  "email": "user@example.com",
  "password": "StrongPassword123"
}
```

Responses

Success (200)

```
{
  "accessToken": "JWT token",
```

```
  “expiresIn”: 30
}
```

Invalid Credentials (401)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 401,
  "error": "Unauthorized",
  "message": "Invalid email or password",
  "path": "/api/v1/auth/login"
}
```

3. Create Store

POST /stores

Request

```
{
  “ownerId”: “uuid”,
  "name": "Ravi Kirana Store",
  "address": "MG Road, Bangalore",
  "phoneNo": "9876543210"
}
```

Responses

Success (201)

```
{
  "storeId": "uuid",
  "message": "Store created"
}
```

Duplicate Store (409)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 409,
  "error": "Conflict",
  "message": "Store with this phone number already exists",
  "path": "/api/v1/stores"
}
```

4. Get Store by ID

GET /stores/{storeId}

Responses

Success (200)

```
{
  "id": "uuid",
  “ownerId”: “uuid”,
  "name": "Ravi Kirana Store",
  "address": "MG Road, Bangalore",
  "phoneNo": "9876543210",
  "isActive": true
}
```

Store Not Found (404)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Store not found",
  "path": "/api/v1/stores/{storeId}"
}
```

5. Create Product

POST /products

Request

```
{
  "name": "Colgate Toothpaste",
  "brand": "Colgate",
  "category": "PERSONAL_CARE",
  "mrp": 60.00,
  “description”: “No. 1 toothpaste in India”
}
```

Responses

Success (201)

```
{
  "productId": "uuid",
  "message": "Product created"
}
```

Invalid Input (400)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid product data",
  "path": "/api/v1/products"
}
```

6. Get Products

GET /products

Query Parameters

page = integer (default: 0)
size = integer (default: 20)
brand = string
category = string

Responses

Success (200)

```
{
  "content": [
    {
      "id": "uuid",
      "name": "Colgate Toothpaste",
      "brand": "Colgate",
      "category": "PERSONAL_CARE",
      "mrp": 60.00,
      “description”: “No. 1 toothpaste in India”
    }
  ],
```

```
  "page": 0,
  "size": 20,
  "totalElements": 100
}
```


Invalid Pagination Parameters (400)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid pagination parameters",
  "path": "/api/v1/products"
}
```


Unauthorized (401)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 401,
  "error": "Unauthorized",
  "message": "Authentication required",
  "path": "/api/v1/products"
}
```


7. Add to Inventory

POST /inventory/{storeId}

Request


```
{
  "productId": "uuid",
  "quantity": 50,
  "sellingPrice": 55.00
}
```


Responses

Success (201)


```
{
  "inventoryId": "uuid",
```

```
  "message": "Inventory added"
}
```


Store not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Store not found",
  "path": "/api/v1/inventory/{storeId}"
}
```


Product not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Product not found",
  "path": "/api/v1/inventory/{storeId}"
}
```


8. Update Inventory

PUT /inventory/{storeId}

Request


```
{
  “inventoryId”: “uuid”,
  "quantity": 80,
  "sellingPrice": 53.00
}
```


Responses

Success (200)


```
{
  "inventoryId": "uuid",
  "message": "Inventory updated"
}
```

Store not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Store not found",
  "path": "/api/v1/inventory/{storeId}"
}
```


Inventory not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Inventory not found",
  "path": "/api/v1/inventory/{storeId}"
}
```


9. Get Store Inventory

GET /inventory/{storeId}

Responses

Success (200)


```
{
  "content": [
    {
      "productId": "uuid",
      "productName": "Colgate Toothpaste",
      "quantity": 50,
      "sellingPrice": 55.00
    }
  ],
  "page": 0,
  "size": 20,
  "totalElements": 100
}
```

Store not found (404)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Store not found",
  "path": "/api/v1/inventory/{storeId}"
}
```

10. Create Transaction

POST /transactions

Request

```
{
  “storeId”: “uuid”,
  "type": "CREDIT",
  "currency": "INR",
  "customerPhone": "9876543210",
  "customerName": "Amit",
  “amountAdjustment”: 10.00,
  “category”: “SALE”,
  "items": [
    {
      "productId": "uuid",
      "quantity": 2,
      "unitPrice": 55.00,
      "discount": 5.00
    }
  ]
}
```

Responses

Success (201)

```
{
  "transactionId": "uuid",
  "netAmount": 105.00,
  "category": "SALE",
```

```
  "createdAt": "2026-02-15T10:30:00Z"
}
```


Insufficient Stock (400)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Insufficient stock for product",
  "path": "/api/v1/transactions"
}
```


Store not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Store not found",
  "path": "/api/v1/transactions"
}
```


11. Get Transaction by ID

GET /transactions/{transactionId}

Responses

Success (200)


```
{
  "id": "uuid",
  "type": "debit",
  "currency": "INR",
  "netAmount": 105.00,
  "status": "sale",
  "customerPhone": "9876543210",
  "customerName": "Amit",
  "items": [
    {
      "productId": "uuid",
      "quantity": 2,
```

```
      "unitPrice": 55.00,
      "discount": 5.00
    }
  ]
}
```


Transaction not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Transaction not found",
  "path": "/api/v1/transactions/{transactionId}"
}
```


12. List Transactions of a store

GET /stores/{storeId}/transactions

Query Parameters

customerPhone = string

page = integer (default: 0)

size = integer (default: 20)

fromDate = date

toDate = date

Responses

Success (200)


```
{
  "content": [
    {
      "id": "uuid",
      "type": "debit",
      "currency": "INR",
      "netAmount": 105.00,
      "status": "sale",
      "customerPhone": "9876543210",
      "customerName": "Amit",
```

```
      "createdAt": "2026-02-15T10:30:00Z"
    }
  ],
  "page": 0,
  "size": 20,
  "totalElements": 25
}
```


Store not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Store not found",
  "path": "/api/v1/stores/{storeId}/transactions"
}
```


Invalid Date Range (400)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "fromDate cannot be after toDate",
  "path": "/api/v1/stores/{storeId}/transactions"
}
```


Invalid Pagination (400)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid pagination parameters",
  "path": "/api/v1/stores/{storeId}/transactions"
}
```


13. Create Refund

POST /refunds

Request

{
  “storeId”: “uuid”,
  "transactionId": "uuid",
  "items": [
    {
      "productId": "uuid1",
      "quantity": 2
    },
    {
      "productId": "uuid2",
      "quantity": 1
    }
  ],
  "description": "Customer returned damaged items"
}

Responses

Success (201)

{
  "refundId": "uuid",
  "message": "Refund processed"
}

Store not found (404)

{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Store not found",
  "path": "/api/v1/refunds"
}

Transaction not found (404)

{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Original transaction not found",

```
  "path": "/api/v1/refunds"
}
```

Product not part of transaction (404)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Product not found in original transaction",
  "path": "/api/v1/refunds"
}
```

Refund quantity exceeds Purchased quantity (400)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Refund quantity exceeds purchased quantity",
  "path": "/api/v1/refunds"
}
```

14. Request Report

POST /reports

Request

```
{
  “storeId”: “uuid”,
  "type": "monthly",
  "fromDate": "2026-01-01",
  "toDate": "2026-01-31"
}
```

Responses

Success (200)

```
{
  "reportId": "uuid",
  "status": "processing"
}
```

Invalid Date Range (400)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid date range",
  "path": "/api/v1/reports"
}
```


15. Get Report

GET /stores/{storeId}/reports/{reportId}

Responses

Success (200)


```
{
  "reportId": "uuid",
  "type": "monthly",
  "totalCredits": 50000,
  "totalDebits": 30000,
  "netFlow": 20000,
  "status": "completed"
}
```


Report not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Report not found",
  "path": "/api/v1/stores/{storeId}/reports/{reportId}"
}
```


16. Assign Store Staff Roles

POST /roles

Request

{
  “storeId”: “uuid”,
  "userId": "uuid",
  "role": "CASHIER"
}

Responses

Success (201)

{
  "userRoleId": "uuid",
  "message": "User role assigned successfully"
}

User not found (404)

{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "User not found",
  "path": "/api/v1/roles"
}

Store not found (404)

{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Store not found",
  "path": "/api/v1/roles"
}

Role already assigned (409)

{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 409,
  "error": "Conflict",
  "message": "User already has a role in this store",

```
  "path": "/api/v1/roles"
}
```

17. Update Staff Role in a Store

PUT /stores/{storeId}/roles

Request

```
{
  “storeRoleId”: “uuid”,
  "role": "ADMIN"
}
```

Responses

Success (200)

```
{
  "storeRoleId": "uuid",
  "message": "User role updated"
}
```

Role entry not found (404)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "User role not found",
  "path": "/api/v1/stores/{storeId}/roles"
}
```

Invalid Role (400)

```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid role value",
  "path": "/api/v1/stores/{storeId}/roles"
}
```

18. Remove Staff from Store

DELETE /stores/{storeId}/roles/{userRoleId}

Responses

Success (200)


```
{
  "message": "User removed from store"
}
```


Role Entry not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "User role not found",
  "path": "/api/v1/stores/{storeId}/roles/{userRoleId}"
}
```


19. Get Staff in a Store

GET /stores/{storeId}/roles

Query Parameters

page = integer (default: 0)

size = integer (default: 20)

role = string

Responses

Success (200)


```
{
  "content": [
    {
      "userRoleId": "uuid",
      "userId": "uuid",
      "name": "Ravi Kumar",
      "email": "ravi@example.com",
      "role": "CASHIER"
    }
  ],
  "page": 0,
```

```
  "size": 20,
  "totalElements": 5
}
```


Store not found (404)


```
{
  "timestamp": "2026-02-15T10:45:00Z",
  "status": 404,
  "error": "Not Found",
  "message": "Store not found",
  "path": "/api/v1/stores/{storeId}/roles"
}
```